Name _____ abc123_____

Homework #3

1. Write the Verilog code for a sequential circuit that counts the following sequence in 4 bits: 0,5,A,4,2,D,7,3,C and then starts over

   Inputs should be 4 bit starting count and the CLK, Outputs should be the 4 bits of the count

   [3:0] Cstart for the initial count and [3:0] Count for the count sequence

   You will have to first design the counter and then implement your design

```
module SequenceCounter(Cstart, CLK, Count);
input[3:0] Cstart;
input CLK;
output reg [3:0] Count;
        always @(posedge CLK)
        begin
        case(Count)
                4'h0: Count <= 4'h5;
                4'h5: Count <= 4'hA;
                4'hA: Count <= 4'h4;
                4'h4: Count <= 4'h2;
                4'h2: Count <= 4'hD;
                4'hD: Count <= 4'h7;
                4'h7: Count <= 4'h3;
                4'h3: Count <= 4'hC;
                4'hC: Count <= 4'h0;
                default: Count <= Cstart;
        endcase
```

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```
        end
endmodule
```

2. Then write a test bench to test it and report any errors in the count

```
module Testbench;
reg [3:0] start_value;
        reg clock;
        wire [3:0] result;
        SequenceCounter counter1(
                .Cstart(start_value),
                .CLK(clock),
                .Count(result)
        );
        initial begin
                start_value = 4'b0000;
                clock = 0;
        end
        always #5 clock = ~clock;
endmodule
```